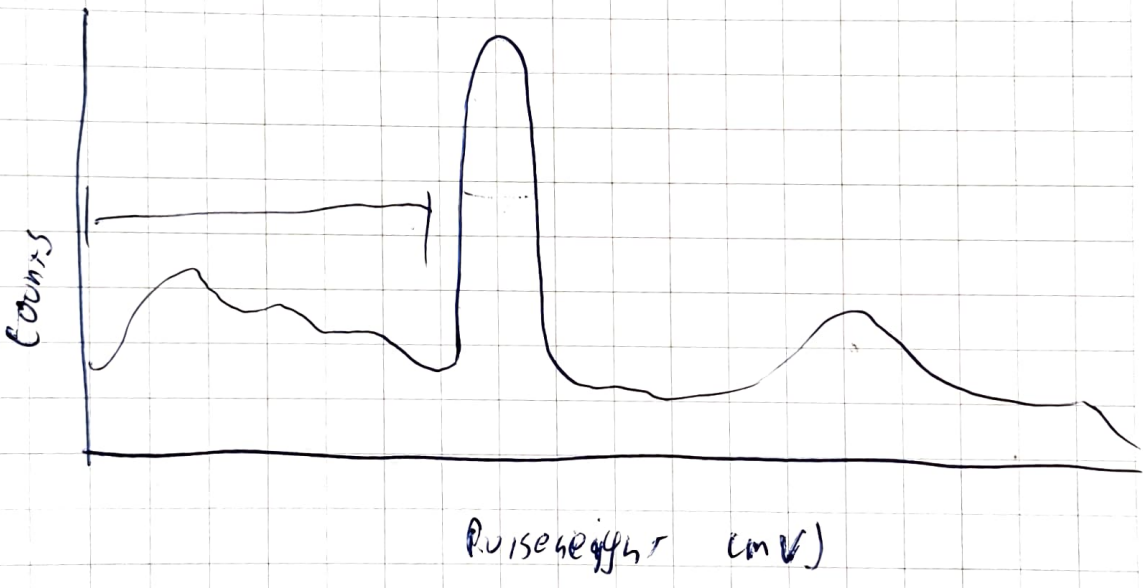
1	Datum: 6-11	Experiment: Particle Physics
	Tijd: 09:50	Onderwerp: Calibration

If we measure radiation we get a value of pulse height. ~~Energy~~ we want to know the energy So we have to find a relation between Energy and Pulseheight.

~~We~~ From our measurements we get a graph ~~for~~ that looks like this



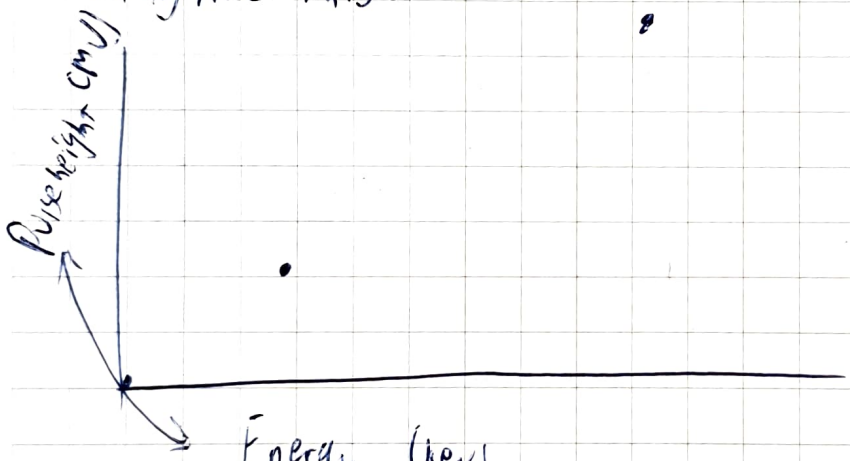
The first part before the peak is known as Comfton Scatering. The radiation loses energy and this explains why the pulseheight is lower. Than the photo peak this is the radiation that hasn't lost energy which we are interested in. For some Isotopes we know at which energy the photo peak is. ~~So~~ If we use these Isotopes we ~~can~~ can find a relation between Pulseheight and Energy. This relation can we use to calculate the energy of unknown radiating sources.

2	Datum: 6-11	Experiment: Particle Physics
	Tijd: 09.53	Onderwerp: Calibration

We know the PhotoPeaks of Na22, Co 60 and Cs 137 these are 510,99, 662 and 1332,5 kev in Python code [Fie Fit.py]

Using a gaussian fit we estimate where the Photo Peaks in Pulse height these are 185,05, 237.8 and 466,8 mV

iAwe Plot thes against each othe we get some thing like this:



now we fit these points together therelation.

Using a linear model ~~a · PulEnergy + b~~
E = a · PulseHeight + b

a = 2,93...
b = ~~-33,95~~ ... 32,53
without Uncertainty

~~now we have to~~

3	Datum: 6-11	Experiment: Particle Physics
	Tijd.: 11:30	Onderwerp: resolution / activity

We have to calculate the resolution using $\frac{\Delta E}{E} \times 100\%$

ΔE = the width of the peak or half the height
And E = the value of the mid Energy.
Calculating the activity you have to know how many photons were emitted.
The detector is at a range of 30,0 m from the source

~~Surface of the detector~~ detector r = 0,03 m detector
Surface$_{det}$ = πr^2 = ~~113,1 cm²~~ 2,827 $m^2 \cdot 10^{-3}$ m
Surface Sphere = $4\pi r^2$ = ~~11,309 cm²~~ 1,130g m^2

~~dee Sur~~ Surfac
Amount of radiation = $\frac{\text{Surface}_{det}}{\text{Surface Sphere}} \cdot 100\% =$
0,25% of the radiation lands on the detector.

There are also photons which lost energy via Compton Scattering to calculate these photons you use the formula:

$N_{Total} = N_{Photopeak} / \frac{\varepsilon_p}{T}$

4	Datum: 6-11	Experiment: Particle Physics
	Tijd: 12:10	Onderwerp: activity

The activity on 1 december 2010 was 301 kBq
So 5817 days have passed since then

$$A_t = A_0 \, e^{\frac{-\ln(2)}{T_{1/2}} t}$$

$T_{1/2}$ of Cs 137 = 30,2 years

t in years = 15,92 years

$$A = 301 \cdot e^{\frac{\ln 2 \cdot 15{,}92}{30{,}2}} = 208{,}82 \text{ kBq}$$

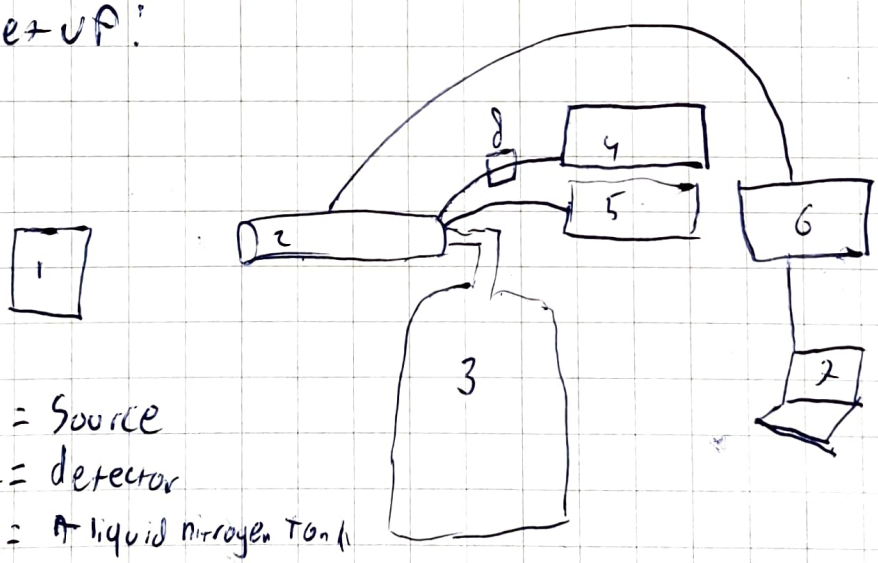
we determined the Resolution of the detector using the formula $\frac{\Delta E}{E} \cdot 100\%$

where ΔE = the width of the peak at half the height and E is the Energy value of the peak.
The resolution is between 8 - 10%

5	Datum: 11-11	Experiment: Particle Physics
	Tijd: 13:15	Onderwerp: Worklion

We wont to investigate with which matter ~~fo~~ the radio active boors and mush rooms are contaminated we wont to dor this by measering the light peak and than with ~~the before~~ calculated calibration we can determine the Energy value of the photo peak. This value conwe compare with known photo peak values of different sources. If we find corresponding photo peaks we know chor the source is contaminated by chor source.

Setup:



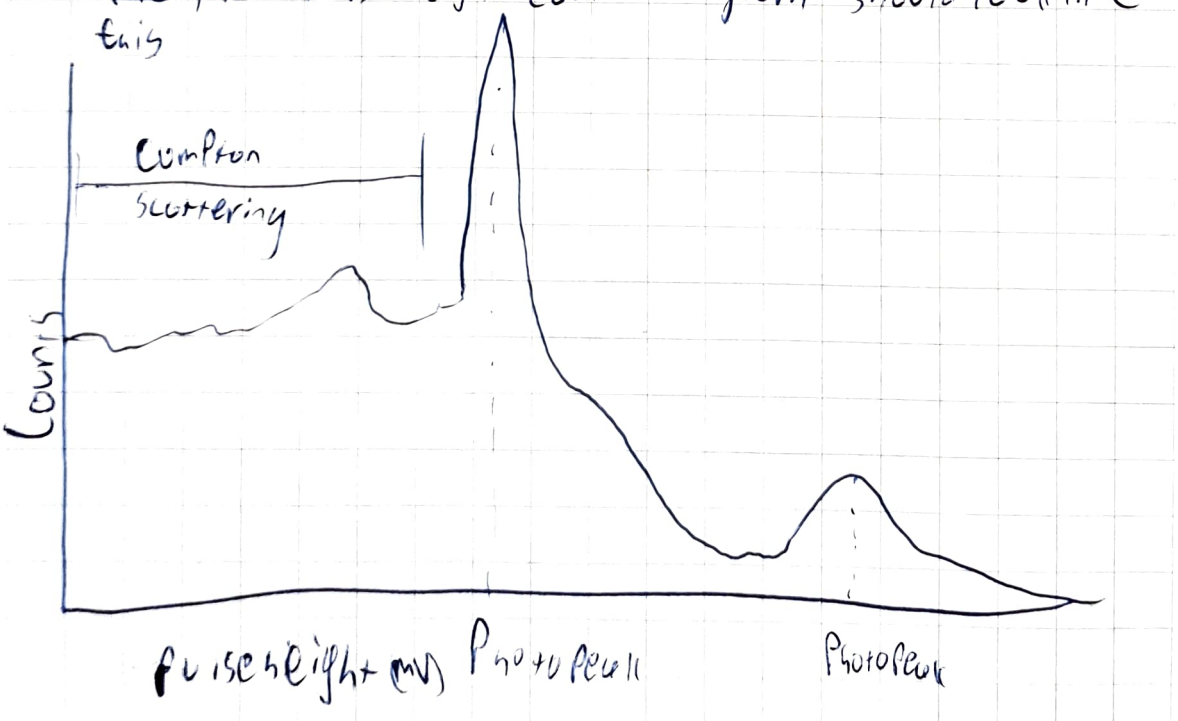
1 = Source
2 = detector
3 = liquid nitrogen Tank
4 = High Voltage
5 = low voltage
6 = Picoscoop
7 = laptop
8 = filter

This is the setup we use we use a ~~Mfode~~ detector which is semiconductor detector with a P-I-N configuration. We use liquid nitrogen to cool the detector. ~~Secondse item~~ the high and low voltage power the detector, the filter ~~kit~~ cuts ~~by~~ voltage peaks so

6	Datum: 11-11	Experiment: Particle Physics
	Tijd: 13:32	Onderwerp: Werkplan setup

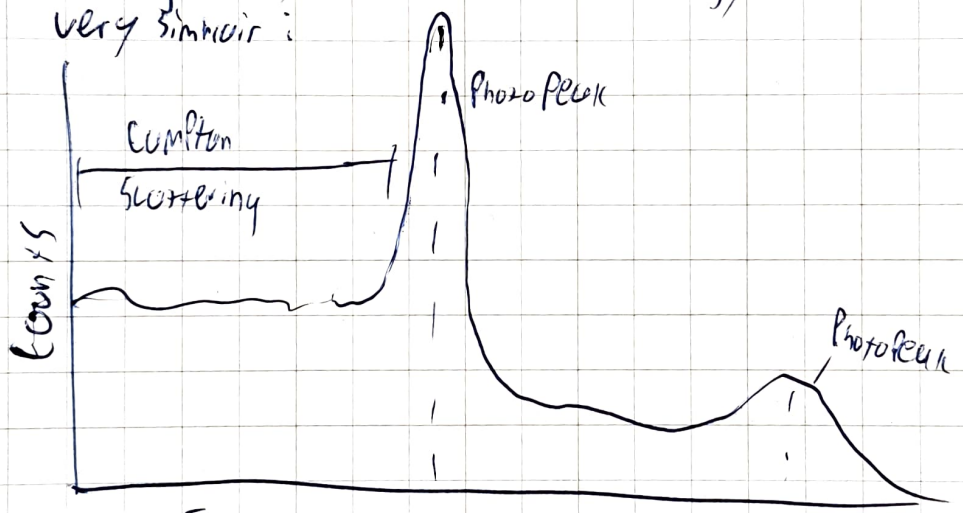
The Oscilloscope (specific the PicoScope 5000) to visualize the data. & The laptop saves the data.
~~the~~ The maximum high voltage = 1000 V
~~Theory~~

Theory
Radioactive sources emit α, β or gamma rays when decaying. ~~The~~ These rays carry energy and this energy we can measure with the detector. ~~If~~ If a ray hits the detector it sends a ~~sig~~ signal with a certain pulse height. ~~The~~
~~To~~ To convert the pulse height to Energy we have to find a relation between pulse height and Energy. To find ~~these~~ ~~the~~ this relation we have to measure the ~~Photo peak~~ of pulse height of sources we know the energy curve of.
~~the~~ The pulse height count graph should look like this



7	Datum: 11-11	Experiment: Particle Physics
	Tijd: 13:54	Onderwerp: Worksheet Theory

the first part of the graph is known as compton scattering. ~~the~~ ~~The roy~~ int ~~the~~ The photon interacted with a particle and lost some Energy before it hit the detector. The photopeak is are all the photons which haven't lost any energy. There can be more than one photopeak that means that there are multiple decays happening. ~~If you look at~~ The Energy count plot look very similair:



Here you can also see the photopeaks and compton scattering.

~~If you want to know~~ We want to know the resolution of the detector we do this with the formula

$$res \quad \frac{\Delta E}{E} \cdot 100\% = \text{resolution}.$$

ΔE is the width of the peak at half the height. and E is the ~~was~~ Energy value of the photopeak

8	Datum: 11-11	Experiment: Particle Physics
	Tijd: 14:00	Onderwerp: Workplan / Theory

We also want to determine the efficiency for the detector relative to the detector previously used.

Experimental workflow / timing:
In session 3-4 we will calibrate the detector to find the relation between the measured pulseheight and the Energy. we do this by measuring the ~~known~~ pulseheight from Na22 Cs137 and Co60 and comparing that with the known Energy of the photopeak. We also want to determine the resolution and efficiency this session.

Session 4
measurements of unknown sources

Session 5
initial conclusion / measurements

Session 6-8
data-analysis, additional measurements interpretation

research question
With which radioactive are the fungi / books contaminated.

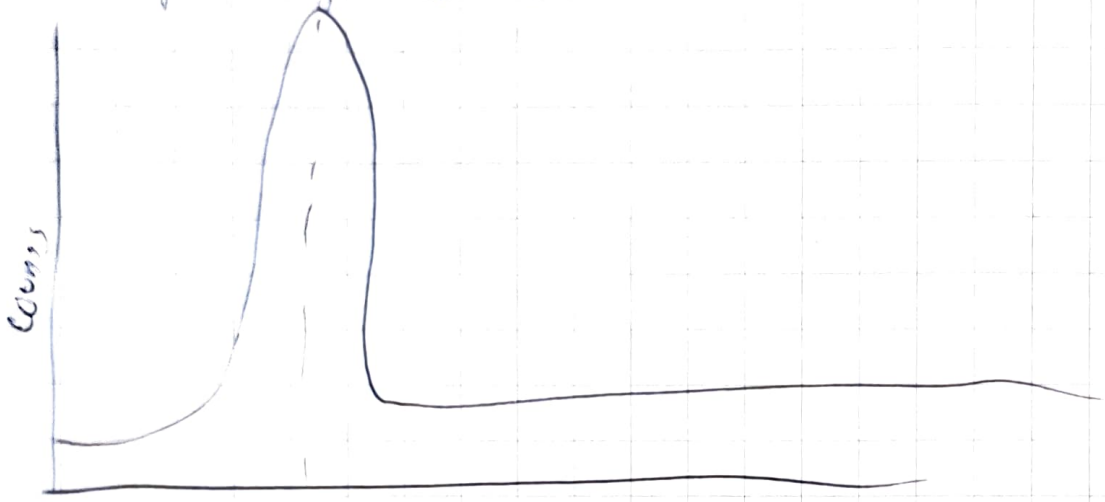
Hypothesis
~~We don't know were the fungi were~~

9	Datum: 11-11	Experiment: Particle Physics
	Tijd: 16:07	Onderwerp: werkplan Hypothesis

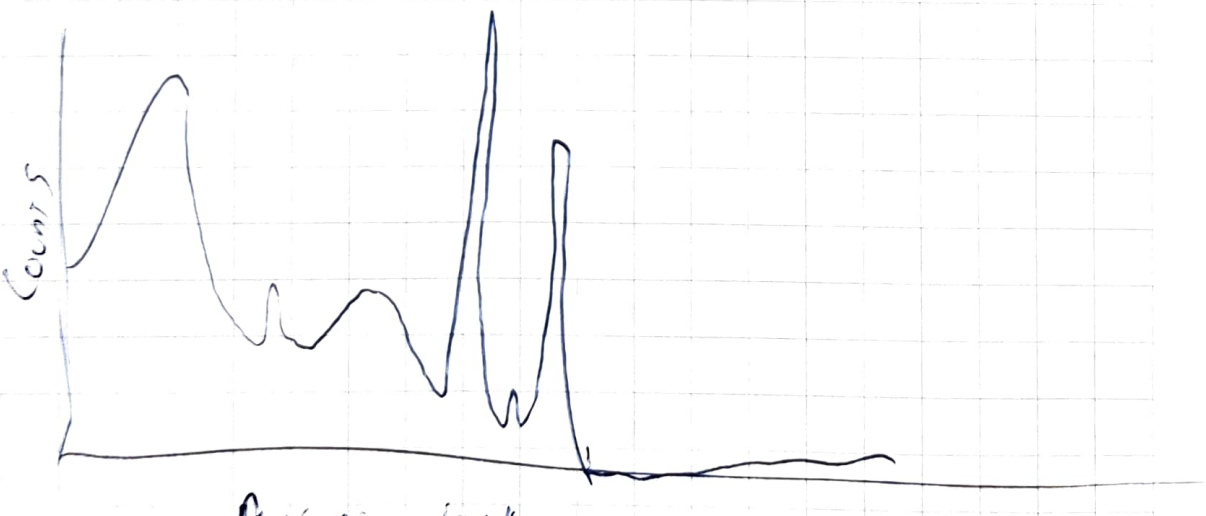
The fungi where ~~contam~~ contaminated due to the Chernobyl explosion. So ~~positive radioactive~~ material such ~~as~~ it is likely that the ~~radium~~ Isotopes Cs_{137}; Cs_{134} or I_{131}. have ~~contaminated~~ contaminated the fungi.
It could also be possible that there is some Sr 90 β and/or Am as well. ~~defo~~ ~~Eu 105~~
~~Short half time so the fungi are likely polluted with Cs 134 or Cs 137, &~~
but these materials where not transported as far as the other elements. As I_{131} has a ~~very~~ short half life so the fungi are likely contained with Cs 137 or Cs_{134}.

10	Datum: 11-11	Experiment: Particle Physics
	Tijd: 15:20	Onderwerp: ~~Metingen~~ Meetmethode

We are measuring the Pulseheight of Cs137
we get a graph that looks like this



There is just one Peak and no compton scattering
Some have to big/it everything right.
with other sources we get the same graph
~~How~~ we change the pulse polarity and move
we get a graph that looks like this



for Co Co

11	Datum: 11-11	Experiment: Particle Physics
	Tijd: 16:40	Onderwerp: Calibration

We have done the measurements with Cs137, ~~and~~ Na22 and Co60 we ~~found~~ used the earlier made code to determine the relation between Pulseheight and Energy.